# 10 · Project sheet · Be a Sorting Robot

*Robot / Maker Lab practical · Nursery 1 · Term 2 · Week 6*

## 1. At a glance

| | |
|---|---|
| **Time** | 40 minutes (fits the lesson; 15 min if used on its own) |
| **Group size** | Groups of 4–6; groups guess each other's rule |
| **Learning goal** | I can sort things by one rule and say my rule. |
| **Coding idea** | A rule decides the groups. Change the rule and the same things make new groups. |
| **Week link** | https://rhinoxscientifica.com/q/n1-t2-w6 |

## 2. Materials needed

| Item | Per | Low-cost swap |
|---|---|---|
| 2–3 hoops | group | Chalk circles or skipping ropes |
| Basket of big objects (nothing smaller than a toilet-roll opening) | group | Picture cards |
| Sorting picture cards | group | Draw them |
| Red, blue, yellow cards | child | Coloured paper |
| Junk boxes + bin labels (optional) | class | Skip |
| Step Star stickers | child | Washable marker star on the hand |

## 3. Safety and care

- Nothing smaller than a toilet-roll tube opening.
- Walk to the hoops.
- Hoops lie flat so no one trips.
- Carry one object at a time.

## 4. Before the lesson

1. Fill the baskets; check every object is big enough.
2. Print the sorting cards and colour cards (Printables).
3. Read the lesson spec the evening before and lay out the cards.

## 5. Step-by-step guide

| Step | Teacher says | Children do |
|---|---|---|
| 1 | "Ayo's basket is mixed up! Let's be sorting robots." | Look at the basket |
| 2 | "Choose a rule. Colour? Shape?" | Group chooses a rule secretly |
| 3 | "Sort everything into the hoops by your rule." | Carry objects to hoops |
| 4 | "Next group: guess the rule!" | Another group guesses |

| Step | Teacher says | Children do |
|---|---|---|
| 5 | “Were they right? Say your rule.” | “Our rule was colour!” |
| 6 | “Now sort the same things by a different rule.” | Sort again |

## 6. Explore it

| | |
|---|---|
| **Question** | Can the same things be sorted in different ways? |
| **Predict** | “If we sort by shape instead of colour, will the groups change?” |
| **Try** | Sort by colour, then sort the same things by shape. |
| **Observe** | “Which things moved to a new group?” |
| **Explain** | The rule decides the groups. A new rule makes new groups. |

## 7. Talk about it

**1a.** What was your rule?

**1b.** Which group has the most?

**1c.** Did the other group guess your rule?

**2a.** What is the same about these things?

**2b.** What is different?

**2c.** Can you sort them another way?

## 8. Support and stretch

| | |
|---|---|
| **Support** | Two groups by colour; the teacher puts the first item in each hoop. |
| **Stretch (optional)** | Sort by two features (colour and shape). Never assessed. |

## 9. How we check

**Observation only, not marked.**

| Look for | Secure | With prompts | Not yet |
|---|---|---|---|
| Sorts objects by one feature | ☐ | ☐ | ☐ |
| Says the rule | ☐ | ☐ | ☐ |
| Guesses another group’s rule | ☐ | ☐ | ☐ |

## 10. Take-home and parent note

> **Dear parent or carer**
> Today we were **sorting robots**: we put things into groups by one rule.
> At home: sort the washing or the shopping and ask “What is our rule?”

## 11. Show and share

- Each group shows its sorted hoops; the class guesses the rule.

- Everyone says: “Sorting is smart!”
- Every child who sorted by a rule and said it gets a **Step Star sticker**.



*The Step Star is a weekly sticker, not a badge. The term badge “Step Thinker · Term 2” is awarded in Week 13.*